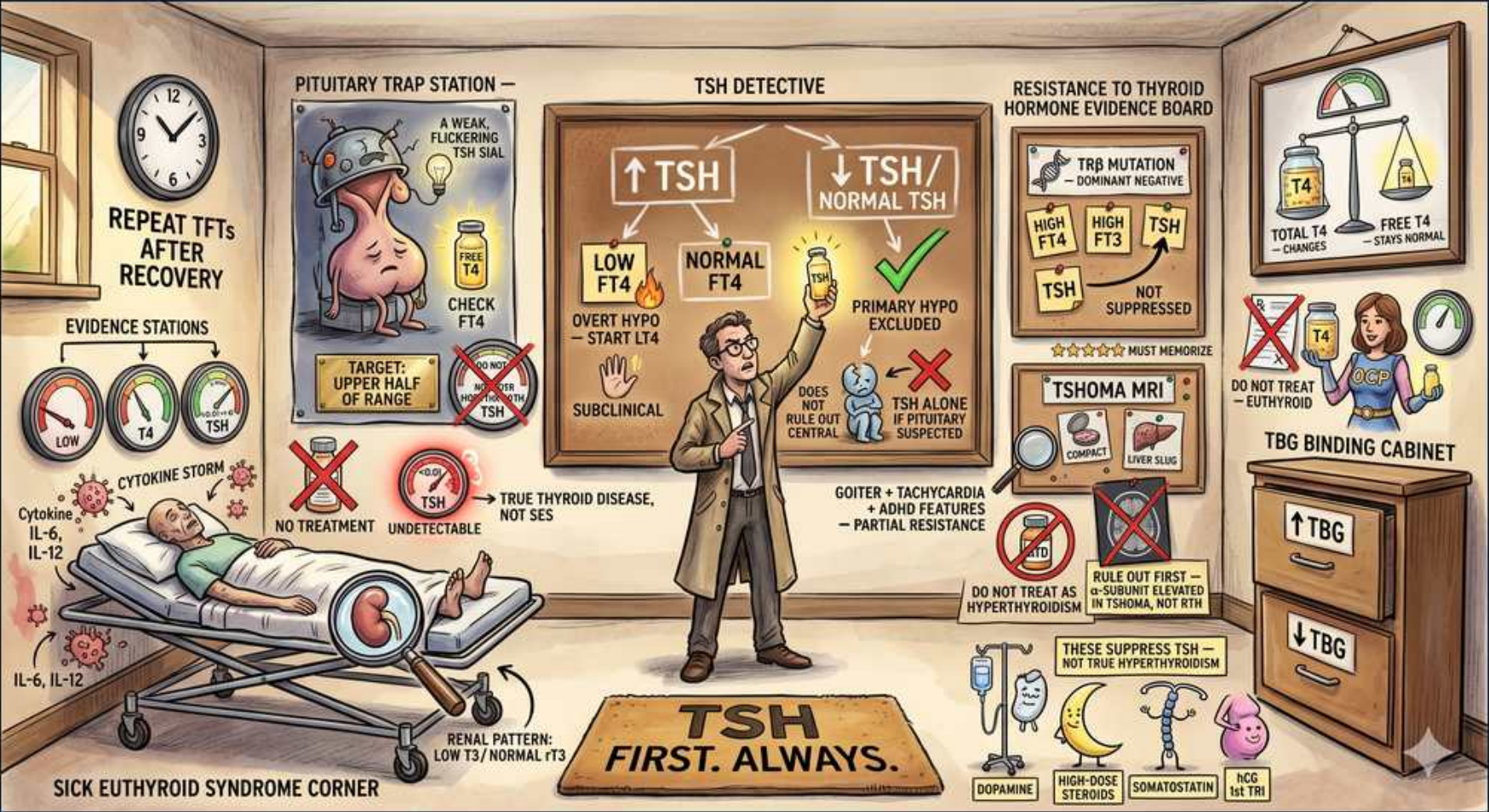


Thyroid — Diagnostic Algorithm



1. TSH FIRST, ALWAYS

WHAT YOU SEE

Doormat 'TSH FIRST. ALWAYS'

WHAT IT ENCODES

TSH is the single best initial screening test for thyroid dysfunction because the pituitary–thyroid axis is a log-linear feedback loop: a small change in free T4 produces a large, amplified change in TSH, so TSH becomes abnormal before free T4 leaves the reference range. Normal TSH ($\approx 0.4\text{--}4.0$ mIU/L) in an outpatient with an intact axis effectively excludes primary thyroid disease. Always pair an abnormal TSH with a free T4 to stage the disorder.

BOARD TRAP

Ordering free T4 or T3 first as the screen — wrong because TSH is more sensitive (amplified) and a normal free T4 with an abnormal TSH would be missed; the board answer is always 'check TSH first.' Exception to remember: if central (pituitary/hypothalamic) disease is suspected, TSH alone is unreliable and you must order free T4 with it.

2. High TSH branch

WHAT YOU SEE

↑TSH arrow on detective's flowchart

WHAT IT ENCODES

A high TSH points to primary hypothyroidism (failing gland → loss of negative feedback → pituitary drives TSH up). The mandatory next step is a free T4 to separate overt from subclinical disease — never act on TSH alone.

BOARD TRAP

Treating with levothyroxine on a high TSH alone — wrong because you cannot tell overt from subclinical (or a transient post-illness rebound / recovering thyroiditis) without the free T4, and many subclinical TSH 4.5–10 cases do not need treatment.

3. High TSH + low FT4

WHAT YOU SEE

'LOW FT4 → OVERT HYPO'

WHAT IT ENCODES

↑TSH with ↓free T4 = overt primary hypothyroidism. This is the pattern that always warrants levothyroxine replacement regardless of symptoms. Confirm cause (anti-TPO for Hashimoto's in iodine-sufficient areas) but do not delay treatment.

BOARD TRAP

Mistaking overt hypothyroidism for the subclinical pattern and withholding treatment — overt (low FT4) is always treated.

4. High TSH + normal FT4

WHAT YOU SEE

'NORMAL FT4 → SUBCLINICAL'

WHAT IT ENCODES

↑TSH with a normal free T4 = subclinical hypothyroidism. Confirm by repeating in 2–3 months (transient causes: recovering thyroiditis, non-adherence rebound, lab variation). Treat when TSH ≥ 10 , or for TSH 4.5–10 if pregnant/planning pregnancy, anti-TPO positive, symptomatic, or goiter/CV risk.

BOARD TRAP

Calling this overt disease or treating every subclinical case — wrong because mild subclinical (TSH 4.5–10) is often transient and may resolve; the threshold to treat unconditionally is TSH ≥ 10 .

5. Low/normal TSH branch

WHAT YOU SEE

'↓TSH / NORMAL TSH' arrow

WHAT IT ENCODES

A low (suppressed) TSH triggers the hyperthyroidism workup: confirm with free T4 and, if free T4 is normal, free T3 (T3 toxicosis). Once thyrotoxicosis is biochemically confirmed, radioactive iodine uptake (RAIU) sorts the cause — high uptake = Graves'/toxic nodule/MNG (gland overproducing) vs low/absent uptake = thyroiditis, exogenous T4, or iodine load (no new synthesis).

BOARD TRAP

Stopping at a suppressed TSH without an uptake scan — wrong because high-uptake (treat with ATD/RAI) vs low-uptake (self-limited thyroiditis, treated only with beta-blockers) have opposite management.

6. Central hypothyroidism

WHAT YOU SEE

'TSH ALONE if pituitary suspected' / 'DOES NOT RULE OUT CENTRAL'

WHAT IT ENCODES

Central (secondary/tertiary) hypothyroidism gives a low free T4 with a TSH that is low OR inappropriately normal (the failing pituitary cannot mount an appropriate TSH rise). TSH alone cannot exclude it — always order free T4 when pituitary disease is suspected, and screen the rest of the anterior pituitary axis (cortisol/ACTH). Replace cortisol BEFORE thyroid hormone to avoid precipitating adrenal crisis.

BOARD TRAP

Using a normal TSH to rule out central hypothyroidism — wrong because the pituitary defect blunts the TSH response, so a 'normal' TSH with low free T4 is itself the abnormal finding.

7. Sick euthyroid syndrome

WHAT YOU SEE

Sick patient, cytokine storm (IL-6, IL-12), 'LOW T3 / NORMAL rT3'

WHAT IT ENCODES

Non-thyroidal illness (sick euthyroid / low-T3 syndrome): acute illness and cytokines (IL-6) impair peripheral 5'-deiodination, shunting T4 to inactive reverse T3 → low total/free T3 with normal or low TSH and often low T4 in severe cases. It is an adaptive response, not true thyroid disease. Do NOT screen TFTs in the acutely ill unless thyroid disease is strongly suspected.

BOARD TRAP

Treating sick euthyroid with levothyroxine — wrong and potentially harmful; it is a self-resolving adaptation. The board move is to NOT treat and recheck TFTs after the patient recovers.

8. Repeat TFTs after recovery

WHAT YOU SEE

Clock 'REPEAT TFTs AFTER RECOVERY'

WHAT IT ENCODES

After acute illness resolves, repeat the thyroid panel to confirm true baseline status — transient sick-euthyroid and post-illness TSH rebound (TSH can briefly overshoot during recovery) both normalize without treatment.

BOARD TRAP

Acting on TFTs drawn during acute illness or during the recovery TSH rebound rather than waiting for the patient to recover and repeating.

9. Pituitary target range

WHAT YOU SEE

Pituitary 'TARGET: upper half of range'

WHAT IT ENCODES

When treating central hypothyroidism, titrate levothyroxine to a free T4 in the upper half of the reference range — you cannot use TSH because the broken pituitary axis makes TSH uninterpretable. Confirm adrenal sufficiency / give glucocorticoid first.

BOARD TRAP

Titration central hypothyroidism by TSH — wrong because TSH is already low/inappropriate from the pituitary lesion and will mislead dosing; follow free T4 instead.

10. TRb mutation / resistance

WHAT YOU SEE

'TRb MUTATION — DOMINANT NEGATIVE', high FT4/FT3 NOT SUPPRESSED, 'DO NOT TREAT'

WHAT IT ENCODES

Resistance to thyroid hormone (THRB / TR β dominant-negative mutation): high free T4 and free T3 with a TSH that is non-suppressed (normal or even high) — the classic 'inappropriate TSH' pattern shared with TSHoma. Generalized resistance is usually euthyroid clinically and is generally NOT treated; pituitary-predominant resistance can be thyrotoxic.

BOARD TRAP

Treating it as Graves'/primary hyperthyroidism and giving antithyroid drugs or RAI — wrong; the suppressed-TSH rule for hyperthyroidism does not apply because TSH is non-suppressed. Distinguish from TSHoma (which needs an MRI and is a tumor).

11. TSHoma

WHAT YOU SEE

'TSHoma — MRI' pituitary

WHAT IT ENCODES

TSH-secreting pituitary adenoma: high free T4/T3 with an inappropriately normal or high TSH (TSH not suppressed despite high hormones). Order pituitary MRI; supportive clues are an elevated alpha-subunit and high SHBG. Treat the tumor (surgery $\pm$ somatostatin analog), not the thyroid.

BOARD TRAP

Diagnosing primary hyperthyroidism and ablating the thyroid — wrong because the lesion is the pituitary; the unsuppressed TSH is the tip-off that separates TSHoma/resistance from Graves'. Differentiate TSHoma (MRI mass, $\uparrow$ alpha-subunit) from TR β resistance (genetic, no mass).

12. TBG binding cabinet

WHAT YOU SEE

'TBG' filing cabinet $\uparrow/\downarrow$ with TOTAL T4 CHANGES / FREE T4 STAYS NORMAL

WHAT IT ENCODES

Thyroxine-binding globulin (TBG) changes alter TOTAL T4/T3 but leave FREE hormone (and TSH) normal — the patient is euthyroid. $\uparrow$ TBG: estrogen/OCP/pregnancy, tamoxifen, acute hepatitis. $\downarrow$ TBG: androgens, glucocorticoids, nephrotic syndrome, severe illness. Always order FREE hormone levels to bypass binding-protein artifact.

BOARD TRAP

Treating an abnormal TOTAL T4 caused purely by a TBG shift (e.g., starting estrogen) — wrong; the free hormone and TSH are normal, so the patient is euthyroid and needs no treatment.

13. Drugs that suppress TSH

WHAT YOU SEE

Dopamine, high-dose steroids, somatostatin 'THESE SUPPRESS TSH — NOT TRUE HYPERTHYROIDISM'

WHAT IT ENCODES

Dopamine (and dobutamine), high-dose glucocorticoids, and somatostatin analogs directly suppress pituitary TSH secretion without causing thyrotoxicosis — a common ICU pitfall. Free T4 is normal and the patient is clinically euthyroid.

BOARD TRAP

Diagnosing hyperthyroidism from a low TSH in a patient on dopamine/steroids/octreotide — wrong; the suppression is drug-induced. Recheck off the drug rather than starting antithyroid therapy.

14. Goiter + ADHD red herrings

WHAT YOU SEE

'Goiter + tachycardia + ADHD features = partial resistance'

WHAT IT ENCODES

Generalized thyroid hormone resistance can present with a mixed clinical picture — goiter, tachycardia, attention/hyperactivity (ADHD-like) features — because different tissues vary in residual receptor sensitivity, so some look hyper and some hypo. The biochemical signature (high FT4/FT3 with non-suppressed TSH) is what reveals it.

BOARD TRAP

Chasing the mixed clinical picture and labeling it primary hyper- or hypothyroidism instead of recognizing the inappropriate-TSH lab pattern of resistance.